

FINAL YEAR RESEARCH PROJECT

End-Term Presentation



MULTIMODAL DEPRESSION DETECTION USING SPEECH AND LINGUISTIC BEHAVIOUR ANALYSIS ACROSS CONVERSATIONAL PHASES

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Presentation Outline

- Problem Statement & Motivations
 - Problem Statement
 - Motivations
- Literature Review
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- Proposed Methodology & Approach
- Key Components/Modules & Algorithms Used
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Problem Statement

- Depression is a serious and widespread mental health disorder that often remains undiagnosed or untreated.
- Traditional diagnosis relies on clinical assessments, which are time-consuming, subjective, and require expert intervention.
- There is a growing need for automated and scalable systems that can detect depression early by analyzing behavioral signals such as speech patterns and linguistic expressions.

Motivations

- Early detection of depression can help prevent severe psychological and social consequences
- Speech and language patterns provide valuable insights into an individual's mental state
- The availability of conversational datasets enables large-scale automated analysis
- AI-based systems can assist clinicians by providing objective and scalable mental health assessment tools

Table 1: Summary of Existing Approaches and Research Limitations

S. No.	Author & Year	Methods/Techniques Used	Dataset Used	Research Outcomes	Findings
1	Gong & Poellabauer (2017)	Feature Extraction: Topic Modeling (context-aware segmentation) Classification: Multimodal feature vector + feature selection	DAIC-WOZ dataset	Improved performance over baseline methods	Captures temporal/contextual details effectively; limited by small sample size and feature dimensionality issues
2	Park & Moon (2022)	Feature Extraction: BERT-CNN (text), CNN-BiLSTM (audio) Classification: Attention-based multimodal fusion	DAIC-WOZ dataset	Improved accuracy over single-modal models	Attention-based fusion improves accuracy; still dependent on quality of preprocessing and multimodal alignment
3	Guo et al. (2022)	Feature Extraction: Transformer-based (RoBERTa, wav2vec 2.0) with Topic Attention Classification: Two-branch multimodal transformer with late fusion	DAIC-WOZ dataset	Accuracy: 90%+ (reported improvement) Recall: Improved via attention	Effective use of topic-level attention improves detection; handles data scarcity using pre-trained models but still limited by small dataset size

S. No.	Author & Year	Methods/Techniques Used	Dataset Used	Research Outcomes	Findings
4	Yoo & Oh (2023)	Feature Extraction: CNN (audio), Transformer (text) Classification: Tensor Fusion Network + LSTM	DAIC-WOZ dataset	Moderate improvement (no exact metrics provided)	Fusion of text and voice improves detection; lacks detailed quantitative evaluation and scalability analysis
5	Nykoniuk et al. (2025)	Feature Extraction: CNN + Bi-LSTM + Self-Attention Classification: Early and Late Fusion models	DAIC-WOZ, EDAIC-WOZ datasets	Accuracy: 86% F1 Score: 0.79	Early fusion performs better; multimodal improves accuracy but dataset imbalance affects reliability
6	Xu et al. (2025)	Feature Extraction: BERT (text), wav2vec 2.0 (audio) Classification: Bi-LSTM + multi-scale convolution + adaptive pooling	CMDC & DAIC-WOZ datasets	F1 Score: Improved significantly RMSE reduced	Strong cross-modal feature integration; handles multiple datasets but requires large training data and complex architecture
7	Gan et al. (2025)	Feature Extraction: LLM (LLaMA), Audio processing Classification: Teacher-Student multimodal fusion with attention	DAIC-WOZ dataset	F1 Score: 0.991	Very high accuracy using teacher-student architecture; strong multimodal fusion but computationally complex and data-dependent

Research Gap & Improvements

- Most existing approaches treat the entire interview as a single unit for analysis
- This global representation fails to capture temporal variations in speech and behavioral patterns
- Important changes in emotional state, hesitation, and response patterns across different stages of the conversation are overlooked
- As a result, subtle indicators of depression that emerge gradually may not be effectively detected

Proposed Improvement:

- The proposed system analyzes depression across three conversational phases i.e. beginning, middle, and end of the interview.
- For each phase, the model generates a depression probability score based on both audio and linguistic features.
- These phase-wise scores are then compared to identify in which phase depression indicators are most prominent.
- This allows us to understand how depressive behavior evolves during the conversation and determine the phase where depression is most clearly detectable.

*“We are not just predicting depression,
we are analyzing how depression signals change across different phases of the conversation.”*

Proposed Solution & Architecture

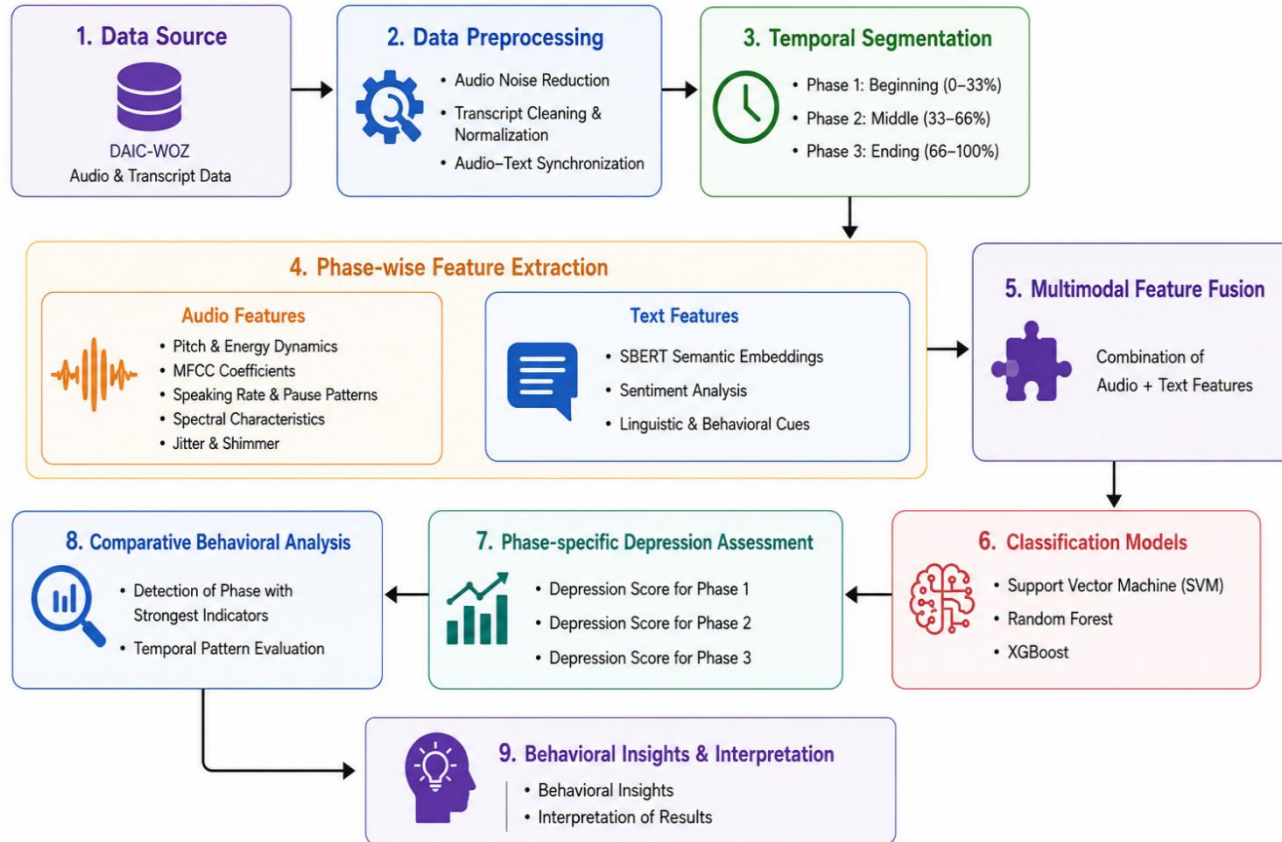


Fig. 2: Schematic Model Diagram of the Proposed Model

Proposed Solution & Architecture cont.

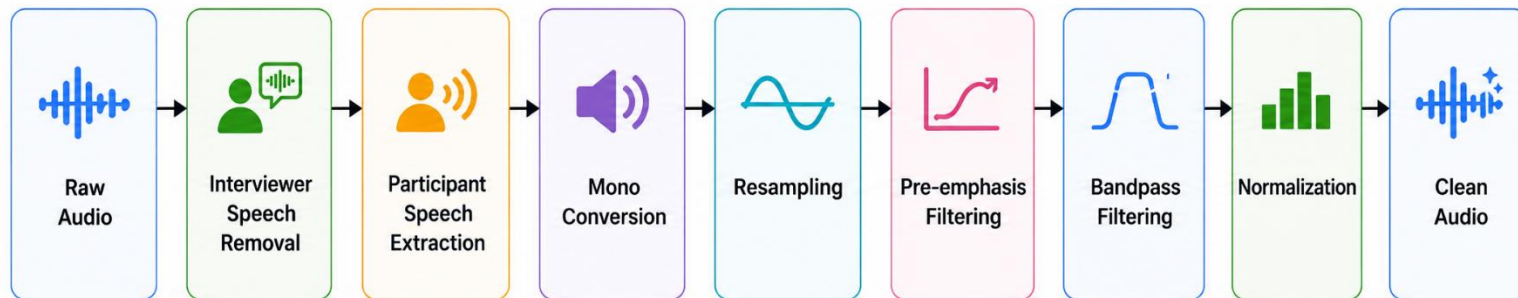


Figure 3. Audio preprocessing pipeline from raw WAV to cleaned participant waveform.

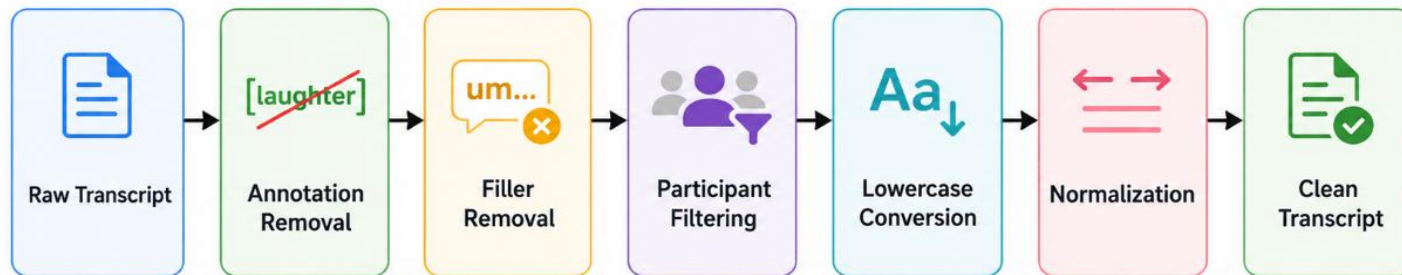


Figure 4. Text preprocessing pipeline from raw transcript to cleaned participant transcript.

Proposed Solution & Architecture cont.

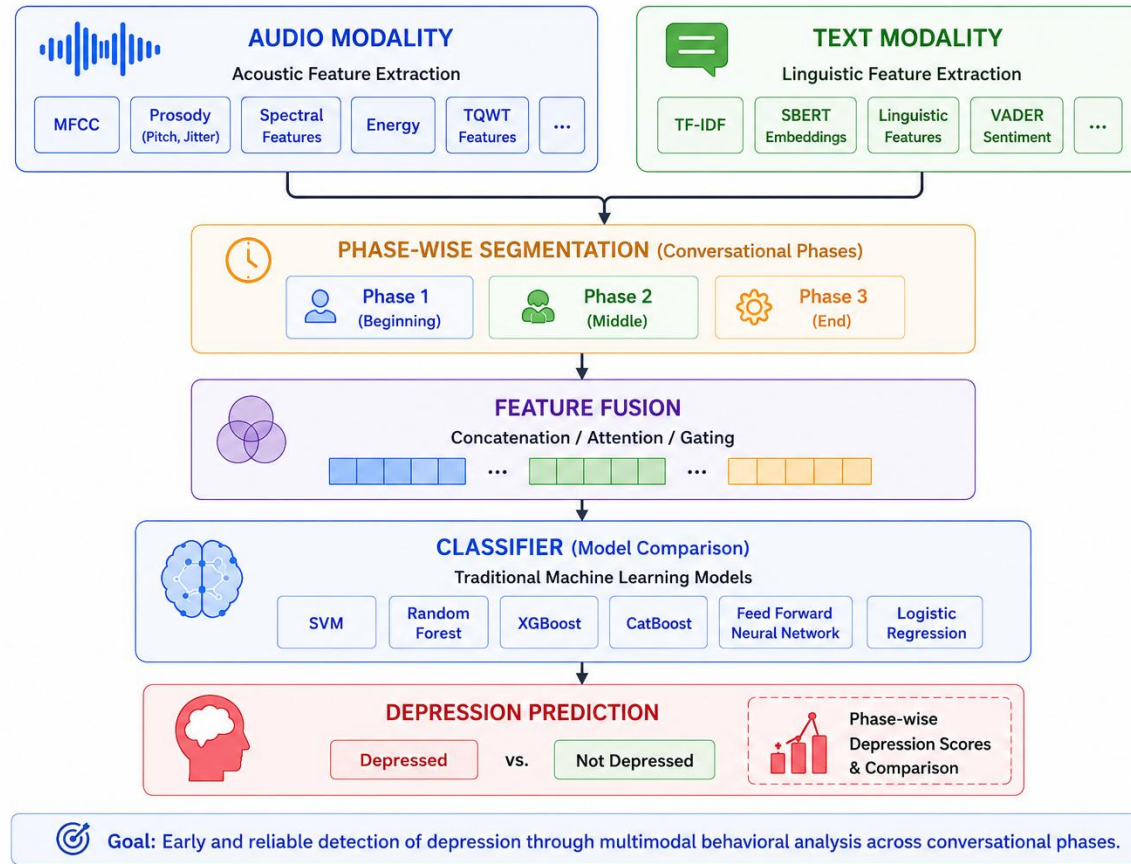


Figure 4. Multimodal Architecture For Depression Detection.



- The proposed system performs multimodal depression detection by analyzing both speech and linguistic behavioural information obtained from conversational clinical interviews. Instead of treating the complete interview as a single unit, the framework performs phase-wise conversational analysis to study how depression-related behavioural patterns evolve across different stages of interaction.
- Initially, multimodal conversational data consisting of participant audio recordings, conversational transcripts, and depression labels are obtained from the DAIC-WOZ dataset. Both audio and textual modalities undergo preprocessing to remove irrelevant information and retain only participant-specific behavioural content.
- For audio preprocessing, interviewer speech is removed, participant speech segments are extracted, and multiple signal enhancement operations such as resampling, filtering, and normalization are applied to improve acoustic consistency. Similarly, transcript preprocessing involves annotation removal, filler-word elimination, participant filtering, lowercase conversion, and text normalization to obtain clean linguistic data suitable for analysis.

- After preprocessing, the complete interview is divided into three conversational phases:
 - Beginning Phase
 - Middle Phase
 - Ending Phase
- This phase-wise segmentation enables temporal behavioural analysis by examining how emotional and conversational characteristics vary throughout the interview.
- Feature extraction is then performed separately for both modalities across each conversational phase. From the audio modality, behavioural acoustic features such as MFCC, pitch, prosodic characteristics, spectral features, energy dynamics, pause patterns, and TQWT-based representations are extracted. From the textual modality, semantic and linguistic features including SBERT embeddings, sentiment information, and behavioural linguistic cues are obtained.

- The extracted audio and textual representations are combined using multimodal feature fusion to generate richer behavioural feature representations. These fused multimodal features are then provided to multiple machine learning classifiers including Support Vector Machine (SVM), Random Forest, XGBoost, CatBoost, Feed Forward Neural Network (FNN), and Logistic Regression for comparative depression classification.
- The system generates depression predictions separately for each conversational phase, allowing comparative phase-wise behavioural assessment. This helps identify which conversational stage contains the strongest depression-related indicators and provides deeper understanding of temporal behavioural evolution during clinical interaction.
- Finally, the framework performs behavioural interpretation and comparative analysis based on phase-wise multimodal predictions to determine whether the participant is depressed or non-depressed while also analyzing how depression-related behavioural signals vary across conversational phases.

Dataset & Tools/Technologies

Dataset Used — DAIC-WOZ

- Multimodal clinical interview dataset designed for psychological distress analysis
- Contains participant audio recordings, conversational transcripts, and PHQ-based depression labels
- Provides participant–interviewer interactions suitable for behavioural and emotional analysis
- Used to study depression-related acoustic and linguistic behavioural patterns across conversational phases

Audio Processing & Feature Extraction

- Librosa — MFCC, spectral, chroma, and acoustic feature extraction
- Parselmouth (Praat) — pitch, pauses, and speech behavioural analysis
- TQWT-based analysis for non-stationary emotional speech representation

Text & Linguistic Analysis

- Sentence-BERT (SBERT) — semantic text embeddings
- VADER Sentiment Analysis — sentiment and emotional polarity extraction
- Transcript preprocessing and linguistic behavioural analysis

Machine Learning & Experimentation

- Scikit-learn — SVM, Random Forest, Logistic Regression
- XGBoost & CatBoost — ensemble-based depression classification
- Pandas & NumPy — preprocessing, feature handling, and dataset management

Development Environment

- Python 3.12
- Jupyter Notebook
- Matplotlib & Seaborn for visualization and comparative analysis

Experimentation

System Specifications:

- Hardware Specifications:

Component	Specification
Processor	Intel Core i7-12650H (12th Gen)
RAM	16 GB
GPU	NVIDIA GeForce RTX 3050 (4GB)
Operating System	Windows 11 Home
System Type	64-bit Operating System

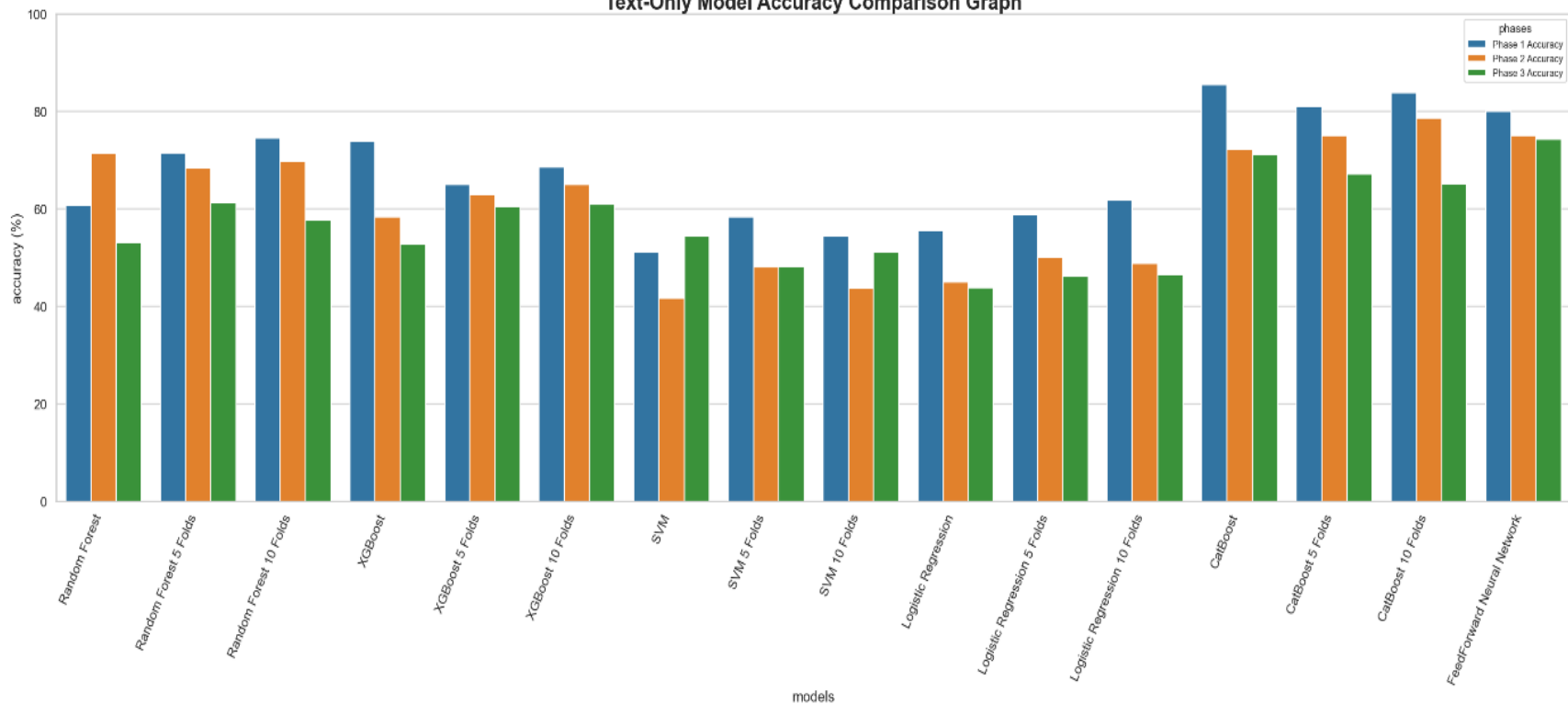
- Software Specifications:

Software / Tool	Purpose
Python 3.12	Core implementation
Jupyter Notebook	Experimentation environment
Librosa	Audio feature extraction
Parselmouth	Speech signal analysis
Sentence-BERT (SBERT)	Semantic text embeddings
Scikit-learn	Machine learning models
Pandas & NumPy	Data handling and preprocessing
Matplotlib & Seaborn	Visualization and plotting

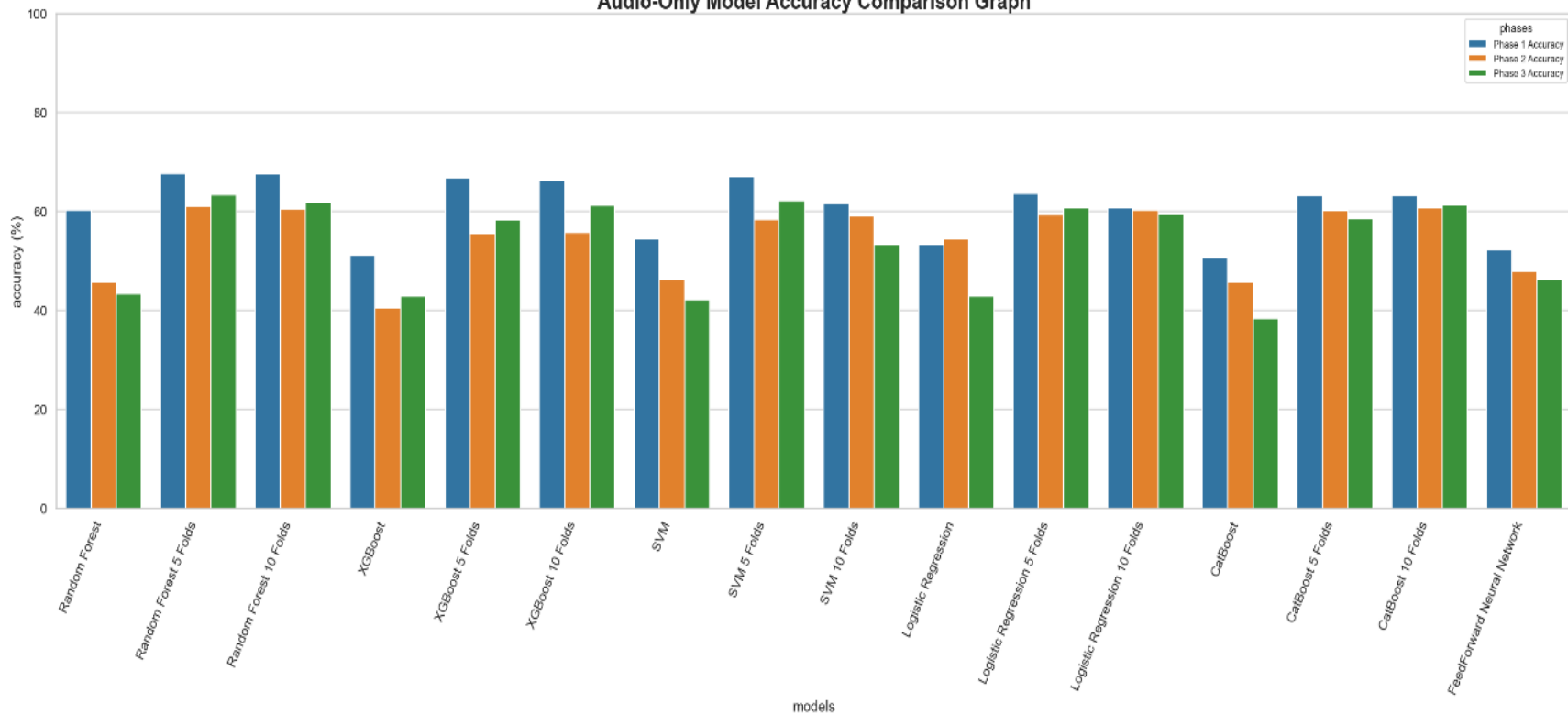
Experimental Analysis:

- The experimental results clearly demonstrate that the multimodal approach consistently achieved better classification performance compared to text-only and audio-only models across all conversational phases.
- By combining both audio and linguistic information, the multimodal framework was able to capture complementary behavioral patterns associated with depression more effectively.
- The text-only models primarily captured semantic and emotional information from participant transcripts, while the audio-only models focused on acoustic characteristics such as pitch, pauses, energy variation, and speaking rate. However, each individual modality contained limited behavioral information when used independently.
- In contrast, the multimodal framework integrated both vocal and linguistic cues, allowing the classifiers to learn richer and more discriminative behavioral representations.
- This fusion significantly improved model robustness, classification stability, and overall prediction capability. Among all conversational phases, Phase I consistently achieved the highest accuracy across most models in the multimodal experiments.
- This indicates that the beginning stage of the interview contains stronger and more distinguishable depression-related cues.

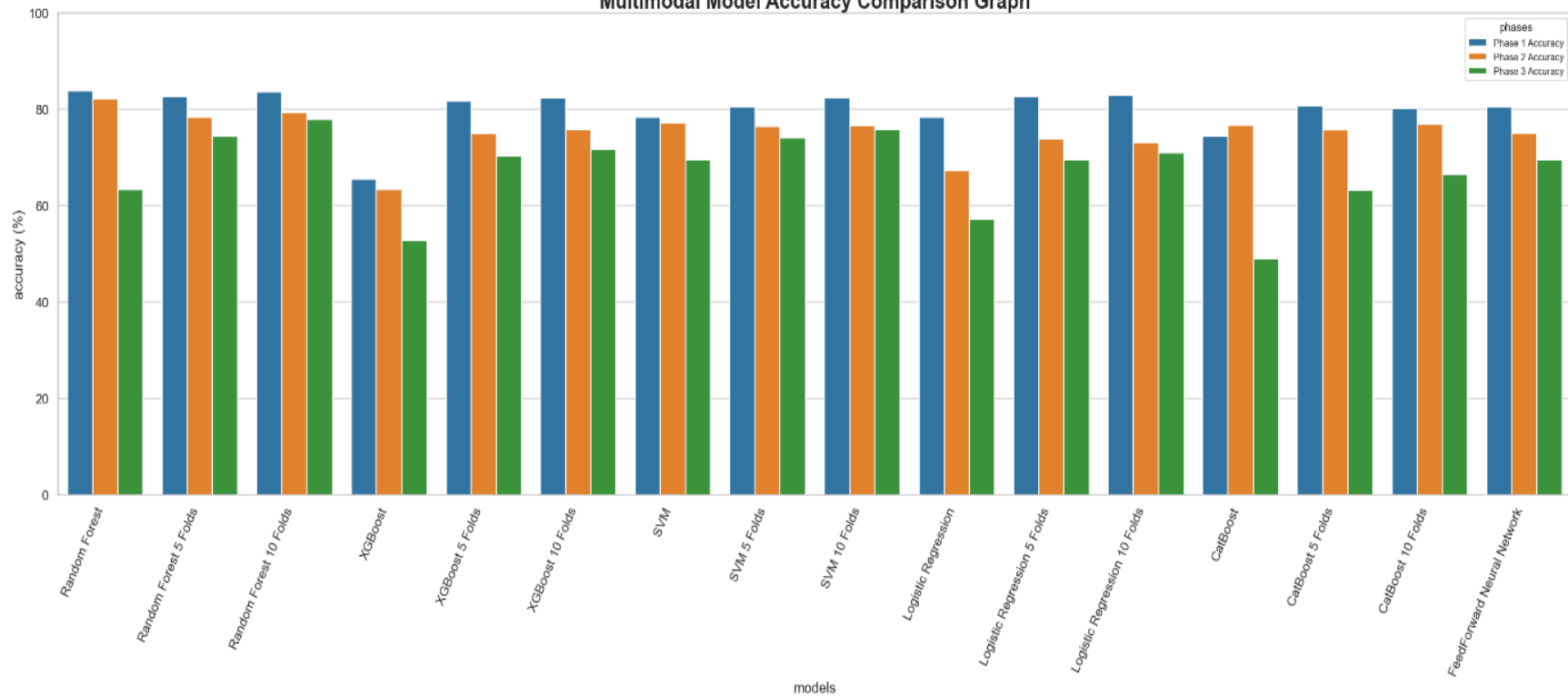
Text-Only Model Accuracy Comparison Graph



Audio-Only Model Accuracy Comparison Graph



Multimodal Model Accuracy Comparison Graph



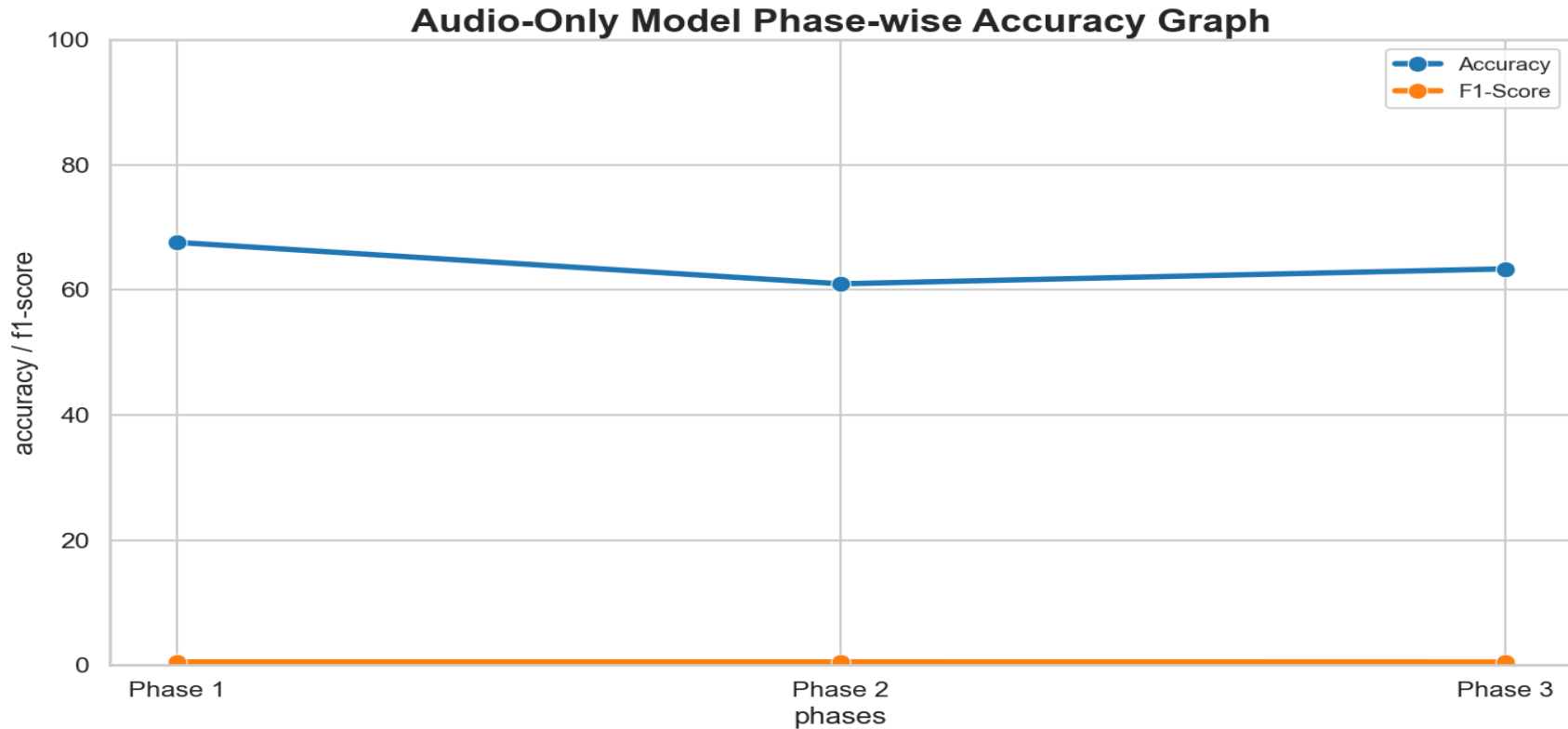
Comparative Performance Analysis of Multimodal Models:

- The proposed multimodal framework was evaluated using multiple machine learning models across three conversational phases of the DAIC-WOZ interviews. Performance was measured using Accuracy, Precision, Recall, and F1-score.
- Among all models, the Random Forest classifier achieved the best overall performance in Phase I with an accuracy of 83.89% and recall of 0.8889, indicating strong depression detection capability during the initial conversational stage.

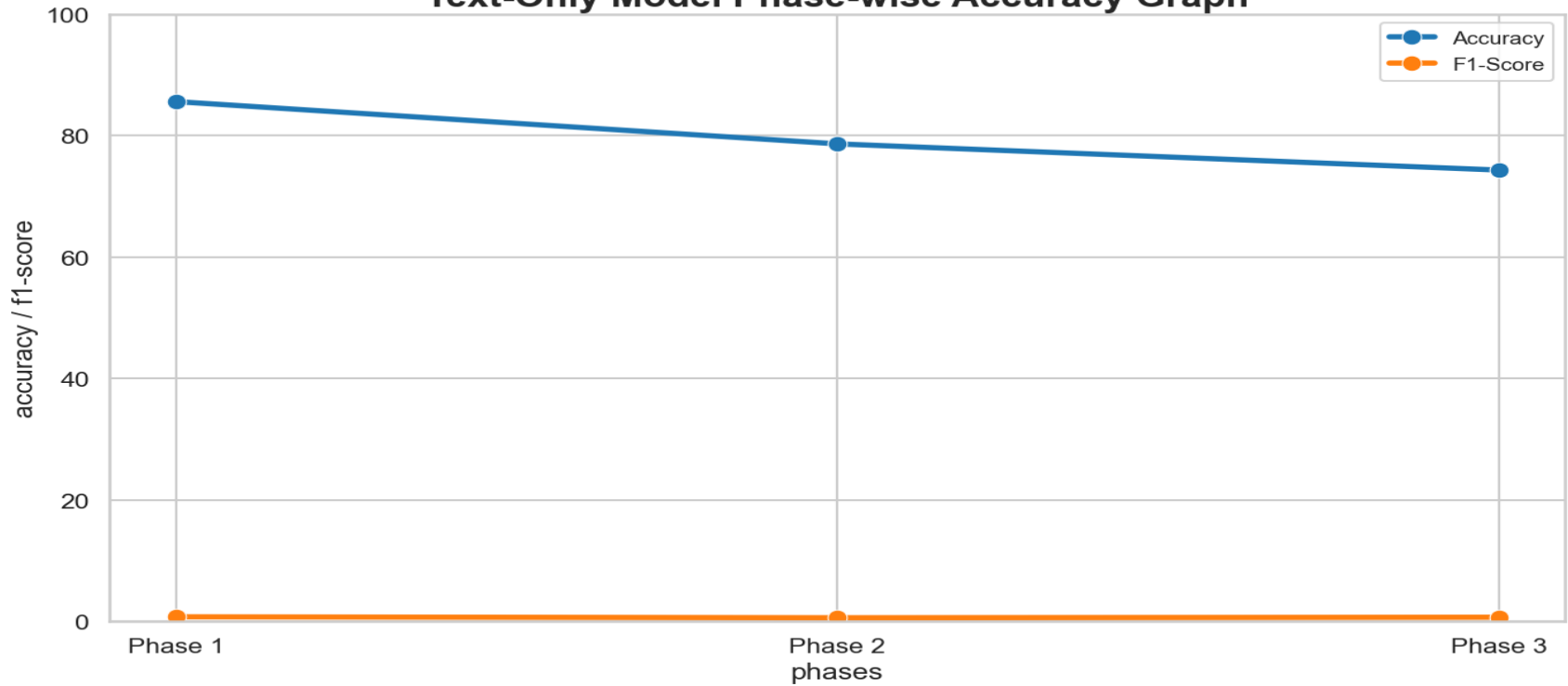
Phase-wise Multimodal Model Performance Comparison

Model	Phase 1				Phase 2				Phase 3			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
Random Forest	83.89%	0.5714	0.8889	0.6957	82.22%	0.5385	0.7778	0.6364	63.33%	0.375	0.3333	0.3529
Random Forest 5 Folds	82.76%	0.5882	0.7143	0.6452	78.43%	0.6486	0.5714	0.6076	74.43%	0.5758	0.4524	0.5067
Random Forest 10 Folds	83.60%	0.6471	0.7857	0.7097	79.29%	0.6316	0.5714	0.6	77.85%	0.6562	0.5	0.5676
XGBoost	65.56%	0.4615	0.6667	0.5455	63.33%	0.5	0.5556	0.5263	52.78%	0.3333	0.3333	0.3333
XGBoost 5 Folds	81.75%	0.5538	0.8571	0.6729	74.96%	0.4909	0.6429	0.5567	70.37%	0.4143	0.6905	0.5179
XGBoost 10 Folds	82.32%	0.5556	0.8333	0.6667	75.88%	0.4808	0.5952	0.5319	71.69%	0.4516	0.6667	0.5385
SVM	78.33%	0.5	0.5556	0.5263	77.22%	0.5556	0.5556	0.5556	69.44%	0.5556	0.5556	0.5556
SVM 5 Folds	80.54%	0.5909	0.619	0.6047	76.48%	0.5532	0.619	0.5843	73.98%	0.5306	0.619	0.5714
SVM 10 Folds	82.49%	0.617	0.6905	0.6517	76.62%	0.5208	0.5952	0.5952	75.76%	0.5778	0.619	0.5977
Logistic Regression	78.33%	0.5833	0.7778	0.6667	67.22%	0.4444	0.4444	0.4444	57.22%	0.3636	0.4444	0.4
Logistic Regression 5 Folds	82.54%	0.5741	0.7381	0.6458	73.79%	0.5435	0.5952	0.5682	69.41%	0.449	0.5238	0.4835
Logistic Regression 10 Folds	82.97%	0.625	0.7143	0.6667	73.02%	0.5	0.619	0.5532	70.95%	0.4902	0.5952	0.5376
CatBoost	74.44%	0.5	0.8889	0.64	76.67%	0.5	0.5556	0.5263	48.89%	0.3333	0.3333	0.3333
CatBoost 5 Folds	80.64%	0.6	0.7857	0.6804	75.73%	0.549	0.6667	0.6022	63.20%	0.3966	0.5476	0.46
CatBoost 10 Folds	80.11%	0.5536	0.7381	0.6327	76.98%	0.551	0.6429	0.5934	66.35%	0.4694	0.5476	0.5055
FeedForward Neural Network	80.56%	0.3913	1	0.5625	75.00%	0.2917	0.7778	0.4242	69.44%	0.3103	1	0.4737

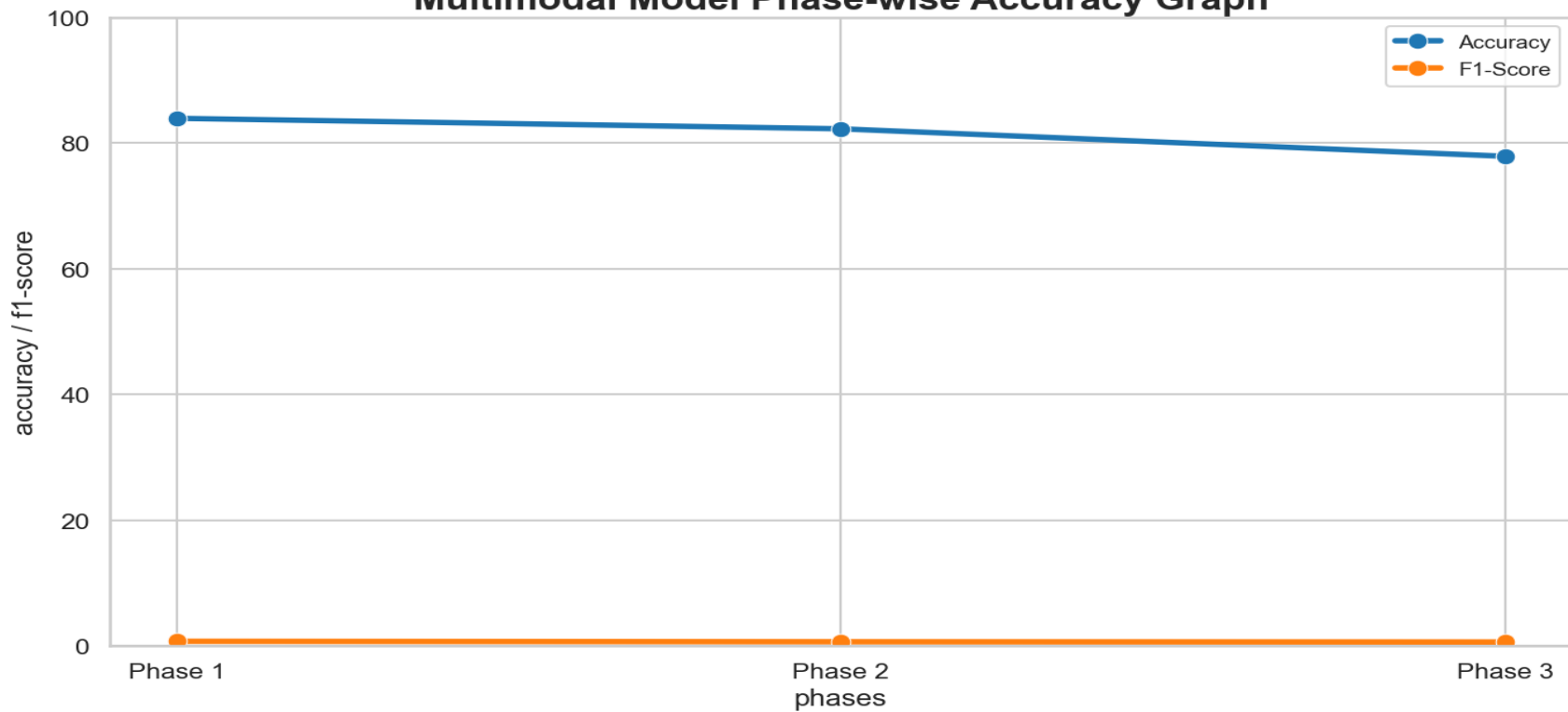
Phase-Wise Depression Analysis:



Text-Only Model Phase-wise Accuracy Graph



Multimodal Model Phase-wise Accuracy Graph



- The phase-wise experimental results show that Phase I consistently achieved the highest classification accuracy across all three approaches.
- In the text-only framework, Phase I produced the strongest performance, indicating that early conversational language patterns contain significant semantic and emotional indicators of depression.
- Similarly, the audio-only framework achieved better performance during the beginning stage, suggesting that initial vocal behavior contains distinguishable acoustic cues such as hesitation, reduced confidence, abnormal pauses, and lower vocal variability.
- The multimodal framework achieved the best overall performance among all approaches by combining audio and textual behavioural information.
- The multimodal model maintained consistently higher accuracy across all phases demonstrating that integrating complementary modalities improves classifications robustness and behavioural understanding.

Performance Analysis:

- The ROC-AUC curves and confusion matrix analyses presented in this study were generated using the proposed multimodal Random Forest classification model.
- The model was trained using fused audio and linguistic features extracted from different conversational phases of the DAIC-WOZ interviews.
- Audio-based behavioral cues such as MFCC, pitch, pauses, and spectral characteristics were combined with text-based semantic and sentiment features to form a unified multimodal feature representation.
- The Random Forest classifier was then evaluated separately on the beginning, middle, and ending conversational phases to analyze how depression-related behavioral patterns vary across different stages of interaction.

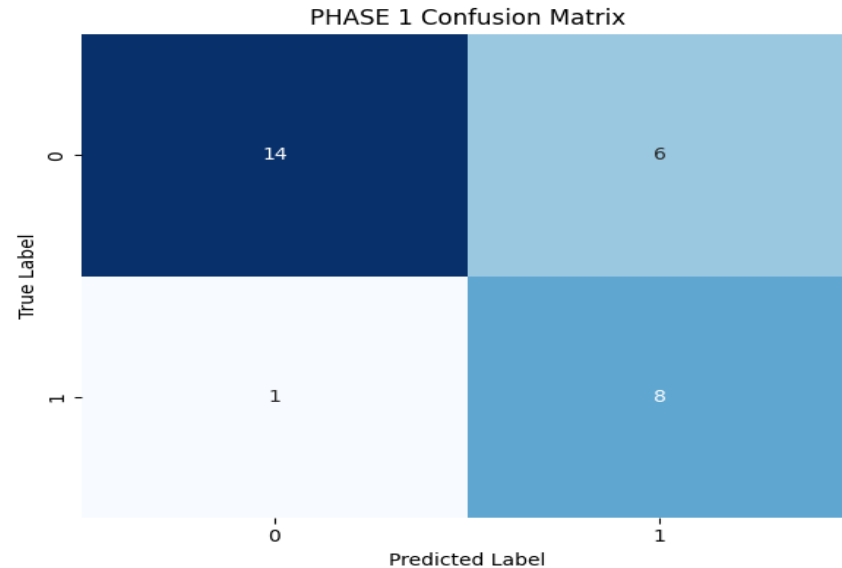
Phase I Analysis :

Phase I achieved the best overall performance among all conversational stages. The model successfully identified most depressed participants with only one false negative, indicating strong sensitivity toward depression-related behavioural cues during the initial interaction stage.

This suggests that early conversational behaviour may contain strong indicators such as:

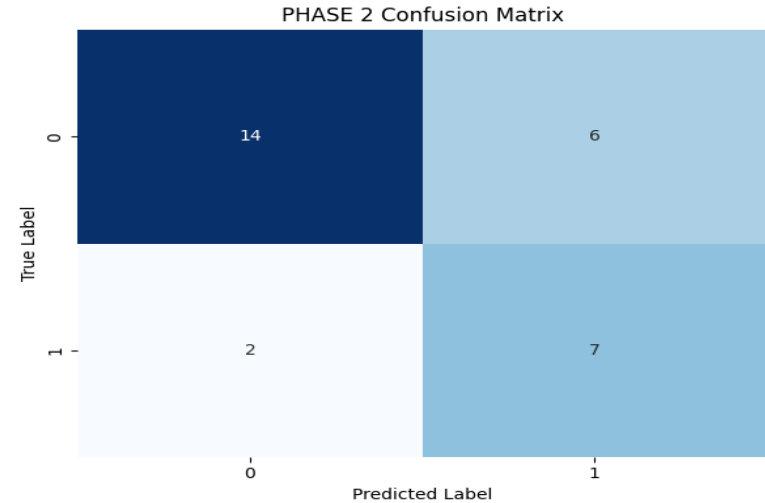
- hesitation
- reduced confidence
- constrained emotional expression
- abnormal speech characteristics

which help improve classification performance.



Phase 2 Analysis :

Phase 2 maintained relatively stable classification performance but showed a slight reduction in depression detection capability compared to Phase I. The increase in false negatives suggests that some depression-related behavioral patterns became less distinguishable during the middle stage of the conversation.



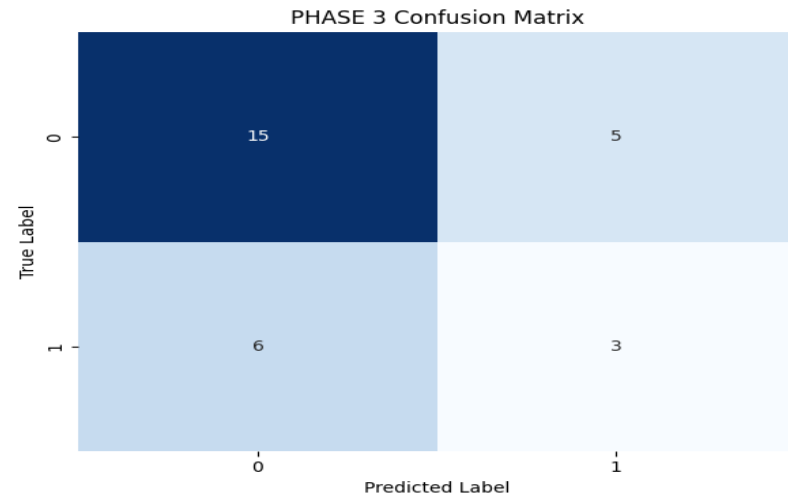
Phase 3 Analysis :

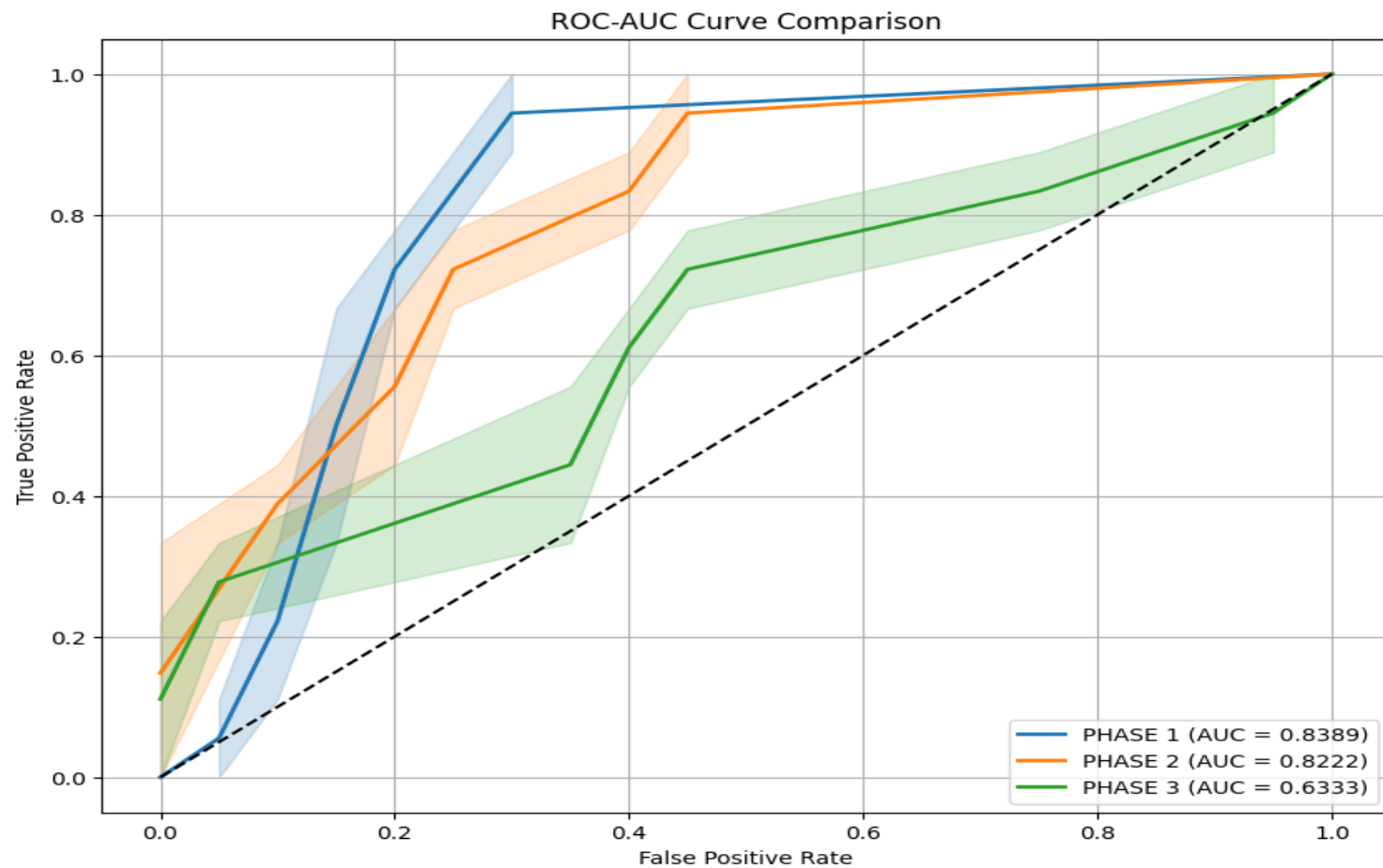
Phase 3 achieved the weakest depression classification performance. Although the model correctly identified most non-depressed participants, it failed to detect several depressed participants, resulting in a higher number of false negatives.

This indicates that depression-related acoustic and linguistic cues became less discriminative during the later stage of the interview.

Possible reasons include:

- conversational adaptation
- emotional stabilization
- reduced behavioural variability
- conversational fatigue





The experimental results show that:

- Phase 1 achieved the highest AUC score of 0.8389, indicating superior classification capability during the initial conversational stage.
- Phase 2 obtained an AUC of 0.8222, showing comparatively strong but slightly lower performance.
- Phase 3 produced the lowest AUC value of 0.6333, suggesting weaker discriminative behavioural patterns during later stages of the conversation.

These findings indicate that depression-related acoustic and linguistic cues were more prominent and distinguishable during the beginning phase of the interview. Initial conversational behaviour such as hesitation, reduced confidence, abnormal pauses, and constrained emotional expression may have contributed to stronger classification performance in Phase 1.

Overall, the ROC analysis validates the effectiveness of phase-wise conversational analysis and demonstrates that early-stage interaction contains significant depression-related behavioural information.

Conclusion and Future Scope

- **Conclusion:**

- A multimodal depression detection framework using speech and linguistic behavior analysis was proposed.
- The interview was divided into beginning, middle, and ending conversational phases to analyze temporal behavioral variations.
- Audio features and textual semantic features were extracted and combined using multimodal fusion.
- Machine learning models such as SVM, Random Forest, and XGBoost were evaluated for depression classification.
- Experimental results showed that multimodal analysis improved classification performance. Phase I achieved the highest accuracy, indicating stronger depression-related cues during the initial stage of conversation.
- The proposed framework effectively captured behavioral patterns associated with depression.

Conclusion and Future Scope

- **Future Scope:**

- Integration of advanced deep learning models such as BiLSTM and Transformers.
- Development of real-time depression monitoring systems.
- Inclusion of additional modalities such as facial expressions and physiological signals.
- Extension to multilingual and cross-cultural datasets.
- Deployment as a clinical decision-support system for mental health applications.

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*Thank
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